

Research Statement

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Markets often fail to price externalities. Pollution, traffic risk, carbon, and the value of ecosystems all carry costs or benefits that no market price reflects. My research measures these externalities and asks who gains and who loses when a policy or a large shock changes them. I combine quasi-experimental methods with Taiwan’s administrative data, including the agricultural census, transit records, accident registries, and high-frequency transaction records. The work falls into two programs. One studies the rural and environmental economy. The other studies the urban and regional economy. Both turn a hard-to-see externality into a number that policymakers can act on.

1. Agricultural and Environmental Economics

Farms must adapt to climate change while contributing to emission reduction, yet they also depend on ecosystem services whose value is rarely reflected in market prices. Governments increasingly intervene through renewable-energy subsidies, food certification systems, and carbon border policies, but credible evidence on whether these interventions change production decisions and environmental outcomes remains limited.

Liou et al. (2025a) examine rooftop solar adoption on Taiwan’s livestock and poultry farms using the 2020 Agricultural Census. Farms that install photovoltaic systems report higher product sales and employ more farm labor, suggesting that energy transition and farm income can complement rather than substitute for one another. Lee et al. (2025a) reach a similar conclusion for chicken farms. Adopters sell about 5.8% more, with the gains arising from higher production volumes rather than price premiums.

Chang and Liou (2026) study food traceability systems, a policy primarily designed to improve food safety. Using plot- and household-level census data together with an instrumental-variable design, the study finds that adoption reduces the probability of applying chemical fertilizer, pesticides, and groundwater by 8.2, 15.4, and 40 percentage points, respectively. At the household level, traceability also increases farm revenue and expands sales through both direct-to-consumer and distributor channels. A policy aimed at information provision therefore generates substantial environmental co-benefits by encouraging less chemical-intensive production.

The remaining studies extend this perspective beyond the farm gate. Liou et al. (2025b) develop a strategic trade model with endogenous green-energy investment to show how the EU Carbon Border Adjustment Mechanism induces cleaner production among non-EU firms while reshaping market competition. Lee et al. (2025b) demonstrate that greater tree diversity not only increases expected forest-farm returns but, more importantly, substantially reduces income volatility, highlighting biodiversity as a productive asset as well as an ecological one. Finally, Chang et al. (2024) identify holiday traffic—not ritual paper burning—as the primary source of the Tomb-Sweeping Day pollution spike, correcting a common attribution error and providing more reliable evidence for designing air-quality policy.

2. Urban, Regional, and Transportation Economics

Moving people generates some of the largest externalities in modern life, including traffic deaths, carbon emissions, and congestion. These effects are hard to identify because transport rules and economic shocks are rarely assigned at random.

Liou et al. (2025c) evaluate the Taipei Metro monthly pass with a difference-in-differences design on station-to-station records. The pass raised ridership by 4.3% and passenger-kilometers by 4.6%, with larger effects on weekends and at higher fares. It lowers net carbon emissions once most of the added trips replace car or motorcycle travel. Liou and Chang (2026a) use a reform in Tainan that removed the hook-turn rule in some townships. A difference-in-discontinuity design shows that accidents fell by 21% and victims by 19%, driven mainly by fewer injuries rather than fewer deaths. Medical spending in the treated areas fell by about 4.7%. Liou and Chang (2026b) study the Dajia Mazu pilgrimage, a multi-day event that moves dense crowds along more than 300 kilometers of public road. Traffic accidents, especially injuries, rise sharply around the event, which gives traffic managers a clear window of elevated risk.

A second set of studies uses the pandemic to see how behavior and regional economies absorb a shock. Chang et al. (2021) provide the first causal estimate of COVID-19 on metro use. Taiwan imposed no lockdown, so the response was voluntary. Each additional confirmed case cut ridership by 1.43%, and the drop was larger at stations near night markets, malls, and colleges. Perceived risk, not mandates, drove the decline. Liou et al. (2024) model how consumers update their beliefs during the pandemic using real-time credit-card data. Spending follows a perfect Bayesian updating pattern. Households spent more on clothing and transportation offline and shifted food spending from offline to online stores. Liou et al. (2026) track every tourist hotel in Taiwan from 2015 to 2024. Revenue and employment recovered unevenly across hotel types and locations, and labor-market rigidity slowed the rebound.

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